

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC Combined Geo-Scientist

Subject: Chemistry

Questions: 132

Source: Official source PDF extraction

AI-generated PYOs: 0

1. [UNKNOWN] paper-ii

1. Consider the following statements regarding radial and total angular function of H-atom:

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

2. [UNKNOWN] paper-ii

1. The radial function $R(r)$ depends on the quantum number and

Official source: backend/data/pvg/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

3. [UNKNOWN] paper-ii

1. $[\text{CoF}_6]^{3-}$ is a paramagnetic and high spin complex

Official source: backend/data/pvg/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

4. [UNKNOWN] paper-ii

1. The low solubility of LiF in water is due to its high lattice enthalpy

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

5. [UNKNOWN] paper-ii

1. It is an outer orbital octahedral complex

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

6. [UNKNOWN] paper-ii

1) (c) ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■

Official source: backend/data/pvg/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

7. [UNKNOWN] paper-ii

1) ((d ■■ =) (1 ■■■■■■■■■■ ===== PAGE 16 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

8. [UNKNOWN] paper-ii

2. The total angular function depends only on the direction and is independent of the distance from the nucleus (r) Which of the statements given above is/are correct? (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2

Official source: backend/data/pyg/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

9. [UNKNOWN] paper-ii

2. Which one of the following is the correct electronic configuration of gold (ground state)? (a) [Xe] 6s2 4f14 5d9 (b) [Xe] 6s1 4f14 5d10 (c) [Xe] 6s2 4f13 5d10 (d) [Xe] 6s2 4f14 5d10

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

10. [UNKNOWN] paper-ii

2. $[\text{Co}(\text{NH}_3)_6]^{3+}$ is a diamagnetic and low spin complex Which of the statements given above is/are correct? (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2

Official source: backend/data/pvg/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

11. [UNKNOWN] paper-ii

2. The low solubility of CsI in water is due to smaller hydration enthalpy of its two ions Which of the statements given

above is/are correct? (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2 ===== PAGE 3 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

12. [UNKNOWN] paper-ii

3. Which one of the following is the correct order of increasing acidic nature of oxides? (a) $\text{SiO}_2 < \text{Al}_2\text{O}_3 < \text{SO}_3 < \text{Cl}_2\text{O}_7$ (b) $\text{SiO}_2 < \text{SO}_3 < \text{Al}_2\text{O}_3 < \text{Cl}_2\text{O}_7$ (c) $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{SO}_3 < \text{Cl}_2\text{O}_7$ (d) $\text{Cl}_2\text{O}_7 < \text{SiO}_2 < \text{SO}_3 < \text{Al}_2\text{O}_3$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

13. [UNKNOWN] paper-ii

3. $\text{trans-}[\text{Co(en)}_2\text{Cl}_2]^+$ Which of the complex ion(s) given above show(s) optical activity? (a) 1 only (b) 1 and 2 (c) 1 and 3 (d) 2 and 3

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

14. [UNKNOWN] paper-ii

4. Which one of the following is the correct outermost electronic configuration of representative elements? (a) $(n-1)d^{1-10} ns^{0-2}$ (b) ns^{1-2} to $ns^{2}np^{1-6}$ (c) $(n-2)f^{1-14} (n-1)d^{0-1} ns^2$ (d) ns^0

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

15. [UNKNOWN] paper-ii

4. See-saw Code: A B C D (a) 4 1 3 2 (b) 2 3 1 4 (c) 2 1 3 4 (d) 4 3 1 2 ===== PAGE 4 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

16. [UNKNOWN] paper-ii

4. It is a diamagnetic complex Which of the statements given above are correct? (a) 1 and 2 only (b) 2 and 3 only (c) 1, 2 and 3 (d) 1, 2 and 4

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

17. [UNKNOWN] paper-ii

4. $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ Which one of the following is the correct order of number of unpaired electrons in the above complexes? (a) $4 > 1 > 2 > 3$ (b) $1 > 2 > 3 > 4$ (c) $3 > 2 > 1 > 4$ (d) $4 > 2 > 1 > 3$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

18. [UNKNOWN] paper-ii

4. $\text{Ba}(\text{OH})_2$ The correct order of basicity of the above hydroxides will be: (a) $\text{Sr}(\text{OH})_2 < \text{Be}(\text{OH})_2 < \text{Ba}(\text{OH})_2 < \text{Mg}(\text{OH})_2$ (b) $\text{Be}(\text{OH})_2 < \text{Mg}(\text{OH})_2 < \text{Ba}(\text{OH})_2 < \text{Sr}(\text{OH})_2$ (c) $\text{Mg}(\text{OH})_2 < \text{Ba}(\text{OH})_2 < \text{Sr}(\text{OH})_2 < \text{Be}(\text{OH})_2$ (d) $\text{Be}(\text{OH})_2 < \text{Mg}(\text{OH})_2 < \text{Sr}(\text{OH})_2 < \text{Ba}(\text{OH})_2$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

19. [UNKNOWN] paper-ii

5. The name of the element given by IUPAC with atomic number 118 is: (a) Nihonium (b) Moscovium (c) Oganesson (d) Tennessine ===== PAGE 2 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

20. [UNKNOWN] paper-ii

6. Which one of the following statements regarding actinoid metals is NOT correct? (a) The actinoid metals are usually silvery in appearance (b) Ionization enthalpies of the early actinoids are higher than that of the early lanthanoids (c) The actinoids are highly reactive metals (d) The actinoid metals in general show +3 oxidation state

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

21. [UNKNOWN] paper-ii

7. Which one of the following is the correct order of ionic radius? (a) $\text{F} > \text{O}^{2-} > \text{Na}^+$ (b) $\text{Na}^+ > \text{F} > \text{O}^{2-}$ (c) $\text{H} > \text{H}^+ > \text{H}^-$ (d) $\text{Al}^{3+} > \text{Mg}^{2+} > \text{N}^{3-}$

22. [UNKNOWN] paper-ii

8. Which one among the following elements has the highest second ionization potential? (a) B (b) Be (c) Mg (d) Al

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

23. [UNKNOWN] paper-ii

9. Consider the following statements regarding magnetism of two complexes:

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

24. [UNKNOWN] paper-ii

10. Consider the following statements in respect of alkali metal halides:

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

25. [UNKNOWN] paper-ii

11. When the radius ratio in an ionic crystal lies between 0.732 – 0.999, the coordination number will be: (a) 3 (b) 4 (c) 6 (d) 8

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

26. [UNKNOWN] paper-ii

12. Which one of the following is the correct order of thermal stabilities of metal carbonates? (a) $\text{BaCO}_3 > \text{CaCO}_3 > \text{MgCO}_3 > \text{BeCO}_3$ (b) $\text{BaCO}_3 > \text{MgCO}_3 > \text{CaCO}_3 > \text{BeCO}_3$ (c) $\text{CaCO}_3 > \text{BaCO}_3 > \text{MgCO}_3 > \text{BeCO}_3$ (d) $\text{BaCO}_3 > \text{BeCO}_3 > \text{MgCO}_3 > \text{CaCO}_3$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

27. [UNKNOWN] paper-ii

13. Which one of the following is the correct order of increasing covalent character of halides of Al^{3+} ? (a) $\text{AlF}_3 < \text{AlCl}_3 < \text{AlBr}_3 < \text{AlI}_3$ (b) $\text{AlI}_3 < \text{AlBr}_3 < \text{AlCl}_3 < \text{AlF}_3$ (c) $\text{AlF}_3 < \text{AlBr}_3 < \text{AlCl}_3 < \text{AlI}_3$ (d) $\text{AlCl}_3 < \text{AlF}_3 < \text{AlBr}_3 < \text{AlI}_3$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

28. [UNKNOWN] paper-ii

14. The appearance of colour in solid alkali metal halides is generally due to: (a) Schottky defect (b) Frenkel defect (c) Interstitial Position (d) F-Centres

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

29. [UNKNOWN] paper-ii

15. Match List-I with List-II and select the correct answer using the code given below the lists: List-I List-II (Molecule/ion) (Shape) A. H_2O

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

30. [UNKNOWN] paper-ii

16. In which of the following molecules/ions, all the bond lengths are NOT equal? (a) BF_3 (b) BF_4 (c) XeF_4 (d) SF_4

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

31. [UNKNOWN] paper-ii

17. Which one of the following exhibits highest bond order? (a) O_2 (b) O_2^- (c) O_2^{2-} (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

32. [UNKNOWN] paper-ii

18. How many resonating structures are possible for NO_3^- ion? (a) 4 (b) 5 (c) 6 (d) 3

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

33. [UNKNOWN] paper-ii

19. Consider the following statements for the complex $[\text{Ni}(\text{NH}_3)_6]\text{Cl}_2$:

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

34. [UNKNOWN] paper-ii

20. Sodium nitroprusside when added to an alkaline solution of sulphide ion gives red coloured solution. The red colouration is due to the formation of: (a) $\text{Na}_4[\text{Fe}(\text{CN})_5(\text{NOS})]$ (b) $\text{Na}_2[\text{Fe}(\text{CN})_5(\text{NO})]$ (c) $\text{Na}_4[\text{FeCl}_5(\text{NOS})]$ (d) $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$ ===== PAGE 5 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

35. [UNKNOWN] paper-ii

21. The complex ions $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are: (a) Both square planar (b) Both diamagnetic (c) Both paramagnetic (d) Paramagnetic and diamagnetic respectively

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

36. [UNKNOWN] paper-ii

23. The denticity of ligands glycinate, iminodiacetic acid and nitrilotriacetic acid are: (a) 2, 3 and 4 respectively (b) 3, 3 and 4 respectively (c) 2, 3 and 3 respectively (d) 1, 2 and 3 respectively

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

37. [UNKNOWN] paper-ii

24. Which one of the following is the IUPAC name of the complex $\text{K}_3[\text{Fe}(\text{CN})_6]$? (a) Potassium hexacyanoferrate(II) (b) Potassium hexacyanoferrate(III) (c) Tripotassium hexacyanoiron(III) (d) Potassium hexacyanoiron(III)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

38. [UNKNOWN] paper-ii

25. Which one of the following complexes will NOT show geometrical isomerism? (a) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ (b) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ (c) $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ (d) $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$ ===== PAGE 6 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

39. [UNKNOWN] paper-ii

26. Which one of the following complexes exhibits optical isomerism? (a) $\text{cis-}[\text{PtCl}_2(\text{NH}_3)_2]$ (b) $\text{trans-}[\text{PtCl}_2(\text{NH}_3)_2]$ (c) $\text{trans-}[\text{Co}(\text{en})_2\text{Cl}_2]^+$ (d) $[\text{Co}(\text{EDTA})]^-$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

40. [UNKNOWN] paper-ii

27. Consider the following complex ions:

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

41. [UNKNOWN] paper-ii

28. A solution is obtained by mixing 10 ml of 0.2 N KOH solution with 30 ml of 0.1 N CH_3COOH . The pH of the solution would be: (Given: pK_a of $\text{CH}_3\text{COOH} = 4.75$; $\log 2 = 0.30$) (a) 5.301 (b) 5.05 (c) 4.699 (d) 4.398

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

42. [UNKNOWN] paper-ii

29. What will be the hydrogen ion concentration of a solution obtained by mixing 500 ml of 0.20 M acetic acid and 500 ml of 0.30 M sodium acetate? (Dissociation constant of acetic acid = 1.75×10^{-5}) (a) $2.75 \times 10^{-5} \text{ mol.dm}^{-3}$ (b) $1.17 \times 10^{-5} \text{ mol.dm}^{-3}$ (c) $3.50 \times 10^{-5} \text{ mol.dm}^{-3}$ (d) $1.75 \times 10^{-3} \text{ mol.dm}^{-3}$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

43. [UNKNOWN] paper-ii

30. Which one of the following represents the correct order of acid strength? (a) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+} < [\text{Al}(\text{H}_2\text{O})_6]^{3+} <$

$\text{Fe}(\text{H}_2\text{O})_6^{3+}$ (b) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+} < \text{Fe}(\text{H}_2\text{O})_6^{3+} < [\text{Al}(\text{H}_2\text{O})_6]^{3+}$ (c) $\text{Fe}(\text{H}_2\text{O})_6^{3+} < [\text{Fe}(\text{H}_2\text{O})_6]^{2+} < [\text{Al}(\text{H}_2\text{O})_6]^{3+}$ (d) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+} < [\text{Al}(\text{H}_2\text{O})_6]^{3+} < [\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ ===== PAGE 7 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

44. [UNKNOWN] paper-ii

32. Which one of the following is the correct order of Lewis Acidity in Boron halides? (a) $\text{BF}_3 < \text{BI}_3 < \text{BBr}_3 < \text{BCl}_3$ (b) $\text{BI}_3 < \text{BF}_3 < \text{BCl}_3 < \text{BBr}_3$ (c) $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3 < \text{BI}_3$ (d) $\text{BBr}_3 < \text{BI}_3 < \text{BF}_3 < \text{BCl}_3$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

45. [UNKNOWN] paper-ii

33. Which one of the following statements regarding acid-base property is NOT correct? (a) $\text{pH} + \text{pOH} = 14$, for all aqueous solutions (b) The pH of 10⁻⁸ (M) HCl solution is 8 (c) The conjugate base of H_2PO_4 is HPO_4^{2-} (d) H_2SO_4 is a strong acid in aqueous solution

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

46. [UNKNOWN] paper-ii

34. What will be the value of pOH of 0.03M aqueous solution of HCl at 25°C? (Given $\log 0.03 = -1.52$) (a) 12.21 (b) 12.48 (c) 11.48 (d) 10.48

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

47. [UNKNOWN] paper-ii

35. What will be the value of mole fraction of the solute in 1.00 molal aqueous solution? (a) 1.7700 (b) 1.7770 (c) 0.0180 (d) 0.0344 ===== PAGE 8 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

48. [UNKNOWN] paper-ii

36. Equivalent weight of $\text{K}_2\text{Cr}_2\text{O}_7$ in acidic medium will be: (Atomic weight of K = 39, Cr = 52 and O = 16) (a) $294 / 2$ (b) $294 / 3$ (c) $294 / 6$ (d) 294

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

49. [UNKNOWN] paper-ii

37. Which one of the following is a primary standard substance? (a) KMnO_4 (b) $\text{K}_2\text{Cr}_2\text{O}_7$ (c) $\text{Na}_2\text{S}_2\text{O}_3$ (d) NaOH

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

50. [UNKNOWN] paper-ii

38. The equivalence point of titration of oxalate solution (acidic medium) with KMnO_4 solution will be: (a) Pink (b) Colourless (c) Orange (d) Light green

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

51. [UNKNOWN] paper-ii

39. In which of the following titrations, an external indicator is NOT necessary? (a) Titration of oxalate solution with KMnO_4 solution (b) Titration of HCl with NaOH solution (c) Titration of Hard water with EDTA (d) Titration of Na_2CO_3 with HCl solution

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

52. [UNKNOWN] paper-ii

40. Which one among the following is a metal ion type indicator? (a) Methylene blue (b) Diphenylamine sulphonic acid (c) Eriochrome black T (d) Cresol red

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

53. [UNKNOWN] paper-ii

41. At what temperature does the total kinetic energy of 0.3 mole of Helium equals the total kinetic energy of 0.4 mole of

Argon at 400 K? (a) 533.3 K (b) 433.3 K (c) 346.3 K (d) 373.0 K ===== PAGE 9 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

54. [UNKNOWN] paper-ii

42. The ratio of Boyle's temperature (TB) and critical temperature (TC) of a gas obeying van der Waals equation is given by: (a) TB : TC = 27: 8 (b) TB : TC = 4: 27 (c) TB : TC = 3: 2 (d) TB : TC = 8: 27

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

55. [UNKNOWN] paper-ii

43. Weight of O₂ necessary to fill up a cylinder of 0.082 litre capacity at 0°C and 96 atm pressure would be: (Compressibility factor = 0.96 and gas constant (R) = 0.082 dm³ atm K⁻¹ mol⁻¹) (a) 11.72 g (b) 1.172 g (c) 117.2 g (d) 10.72 g

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

56. [UNKNOWN] paper-ii

44. Which one of the following gas samples will have the highest average molecular speed? (Atomic weight of O = 16, Ne = 20, C = 12 and He = 4) (a) 1.0 mole of O₂ at 560 K (b) 0.50 mole of Ne at 500 K (c) 0.20 mole of CO₂ at 440 K (d) 2.0 mole of He at 140 K

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

57. [UNKNOWN] paper-ii

45. The root mean square velocity of a monoatomic gas of molar mass 'M' (g mol⁻¹) is 'u'. What is the correct relation between 'u' and average kinetic energy (E) per mole? (a) $u = \sqrt{ME}$ (b) $u = \sqrt{ME/3}$ (c) $u = \sqrt{ME/2}$ (d) $u = \sqrt{ME}$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

58. [UNKNOWN] paper-ii

46. The amount of energy (per mole of a monoatomic gas) that will increase by increasing the temperature by one degree at constant volume is: (a) $C_p - C_v$ (b) $R/2$ (c) $3/2 R$ (d) $C_v + R$ ===== PAGE 10 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

59. [UNKNOWN] paper-ii

47. Which one among the following gases has the largest mean free path under the same external conditions? (a) H₂ (b) Cl₂ (c) N₂ (d) O₂

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

60. [UNKNOWN] paper-ii

48. 50 mL of H₂ gas diffuses through a small hole from a vessel in 20 minutes. Time taken by 40 mL of O₂ gas to diffuse under similar condition will be: (a) 12 minutes (b) 64 minutes (c) 8 minutes (d) 32 minutes

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

61. [UNKNOWN] paper-ii

49. What will be the slope of the plot of log P vs log V for a reversible adiabatic expansion of an ideal gas? [$\gamma = C_p/C_v$] (a) γ (b) $-\gamma$ (c) zero (d) -1

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

62. [UNKNOWN] paper-ii

50. The latent heat of vaporization of Br₂ at 59°C and at 1 atm pressure is 29.2 kJ mol⁻¹. The value of ΔU would be: (R = 8.314×10^{-3} kJ mol⁻¹ K⁻¹) (a) 26.44 kJ mol⁻¹ (b) 2.644 kJ mol⁻¹ (c) 264.4 kJ mol⁻¹ (d) 31.96 kJ mol⁻¹

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

63. [UNKNOWN] paper-ii

51. The ratio of specific heat of gases at constant pressure (C_p) and constant volume (C_v) of N₂ molecule is: (a) 1.667

(b) 1.400 (c) 1.330 (d) 1.520

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

64. [UNKNOWN] paper-ii

52. The total heat content of a system at constant pressure is known as: (a) Enthalpy (b) Entropy (c) Internal energy (d) Activation energy ===== PAGE 11 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

65. [UNKNOWN] paper-ii

53. In the Joule-Thomson expansion of real gas at inversion temperature, the Joule-Thomson coefficient: (a) will be negative if the gas cools post expansion (b) will be positive if the gas gets heated post expansion (c) will be zero if $[\partial(PV)/\partial P]_T$ is positive (d) is zero

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

66. [UNKNOWN] paper-ii

54. For $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, the expression for ΔH is: (a) $\Delta U + 2RT$ (b) $\Delta U + RT$ (c) $\Delta U - 2RT$ (d) $\Delta U - RT$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

67. [UNKNOWN] paper-ii

55. The entropy of vaporization of a liquid is $100 \text{ J mol}^{-1}\text{K}^{-1}$ and it boils at 300K . What would be the value of ΔH_{vap} ? (a) 3.0 kJ mol^{-1} (b) 30.0 kJ mol^{-1} (c) 300 kJ mol^{-1} (d) 3000 kJ mol^{-1}

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

68. [UNKNOWN] paper-ii

56. The maximum efficiency of a Carnot engine operating between 100°C to 200°C will be: (a) 21.85 % (b) 21.14 % (c) 22.14 % (d) 20.14%

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

69. [UNKNOWN] paper-ii

57. Which one of the following is the correct thermodynamic equation of state? (a) $dG = (\partial G/\partial P)_T dP + (\partial G/\partial T)_P dT$ (b) $(\partial U/\partial V)_T = P + T(\partial P/\partial T)_V$ (c) $dU = (\partial U/\partial V)_S dV + (\partial U/\partial S)_V dS$ (d) $T = V - T P, n$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

70. [UNKNOWN] paper-ii

58. Which one of the following changes will necessarily lead to a non-spontaneous change? (a) Positive ΔH and positive ΔS (b) Negative ΔH and negative ΔS (c) Positive ΔH and negative ΔS (d) Negative ΔH and positive ΔS ===== PAGE 12 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

71. [UNKNOWN] paper-ii

59. Equal volumes of what concentrations of $AgNO_3$ and HCl should be mixed to form a $AgCl$ precipitate? [K_{sp} for $AgCl = 10^{-10}$] (a) $10^{-4} \text{ M } AgNO_3$ and $10^{-7} \text{ M } HCl$ (b) $10^{-5} \text{ M } AgNO_3$ and $10^{-6} \text{ M } HCl$ (c) $10^{-6} \text{ M } AgNO_3$ and $10^{-6} \text{ M } HCl$ (d) $10^{-4} \text{ M } AgNO_3$ and $10^{-5} \text{ M } HCl$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

72. [UNKNOWN] paper-ii

60. The solubility product of Ag_2CrO_4 is 32×10^{-12} . What is the concentration of CrO_4^{2-} ion in that solution? (a) $2 \times 10^{-4} \text{ M}$ (b) $16 \times 10^{-4} \text{ M}$ (c) $8 \times 10^{-4} \text{ M}$ (d) $8 \times 10^{-8} \text{ M}$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

73. [UNKNOWN] paper-ii

61. What will be the degree of dissociation of $0.1 \text{ (M) } Ba(NO_3)_2$ solution, if van't Hoff factor of the solution is 2.74? (a)

91.3 % (b) 87 % (c) 100% (d) 74%

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

74. [UNKNOWN] paper-ii

62. Which one of the following colligative properties can provide the most precise molar mass of peptides? (a) Elevation of boiling point (b) Depression of freezing point (c) Relative lowering of vapour pressure (d) Osmotic pressure

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

75. [UNKNOWN] paper-ii

63. At 298 K, the total vapour pressure in Torr for a mixture of volatile components A and B is given by $P = 120 - 75x_B$ (x_B is the mole fraction of B). What is the vapour pressure of pure A and B (in Torr) respectively? (a) 120 and 75 (b) 120 and 195 (c) 120 and 45 (d) 75 and 45 ===== PAGE 13 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

76. [UNKNOWN] paper-ii

64. Which one of the following equations represents the correct relation between relative lowering in vapour pressure and osmotic pressure (π) for a dilute aqueous solution? (a) $\frac{\pi}{P} = \frac{\Delta P}{P}$ (b) $\frac{\pi}{RT} = \frac{\Delta P}{P}$ (c) $\frac{\pi}{T} = \frac{\Delta P}{P}$ (d) $\frac{\pi}{RT} = \frac{\Delta P}{P}$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

77. [UNKNOWN] paper-ii

65. A 1% aqueous solution (mass by volume) of a certain substance is isotonic with a 3% solution of glucose (Molar mass = 180 g mol⁻¹) at a given temperature. The molar mass of the substance (in kg mol⁻¹) will be: (a) 0.60 (b) 180 (c) 0.18 (d) 0.06

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

78. [UNKNOWN] paper-ii

66. Which among the following is the correct order for osmotic pressure of equimolar solutions of glucose, sodium chloride and barium chloride? (a) C₆H₁₂O₆ > BaCl₂ > NaCl (b) C₆H₁₂O₆ > NaCl > BaCl₂ (c) BaCl₂ > NaCl > C₆H₁₂O₆ (d) NaCl > C₆H₁₂O₆ > BaCl₂

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

79. [UNKNOWN] paper-ii

67. 0.05 molal aqueous solution of K₄[Fe(CN)₆] freezes at -0.26°C. Its degree of dissociation will be: ($K_f = 1.86^\circ\text{C Kg mol}^{-1}$) (a) 48.2 % (b) 54.9 % (c) 44.9 % (d) 51.6 % ===== PAGE 14 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

80. [UNKNOWN] paper-ii

68. What will be the value of freezing point depression constant of cyclobutane having freezing point of 280 K and molar enthalpy of fusion of 2.68 kJ mol⁻¹? (Molar mass of cyclobutane = 56.108 g mol⁻¹ and $R = 8.314 \text{ JK}^{-1}\text{mol}^{-1}$) (a) 14.63 K kg mol⁻¹ (b) 16.14 K kg mol⁻¹ (c) 13.64 K kg mol⁻¹ (d) 12.64 K kg mol⁻¹

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

81. [UNKNOWN] paper-ii

69. Which one of the following statements is correct about the Henry's law? (a) It is related to effect of pressure only on solubility of a gas in a liquid (b) It is related to effect of temperature only on solubility of a gas in a liquid (c) It is related to effect of pressure as well as temperature on solubility of a gas in a liquid (d) It is related to effect of pressure as well as temperature on solubility of a gas in a solid

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

82. [UNKNOWN] paper-ii

70. What will be the ratio of mole fractions of O₂ and N₂ dissolved in water at 25°C? (Given that Henry's law constant

(in torr) for O₂ and N₂ are 3.3×10^7 and 6.5×10^7 respectively at 25°C; mole fractions of O₂ and N₂ in air are taken as 0.2 and 0.8 respectively) (a) ~ 0.51 (b) ~0.66 (c) ~0.25 (d) ~0.49

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

83. [UNKNOWN] paper-ii

71. The cell constant with 0.1 N KCl solution is found to be 0.98 cm⁻¹ and the resistance offered by 0.01 N HCl is 89 Ω. The specific conductance of HCl will be: (a) $1.101 \times 10^2 \Omega^{-1} \text{ cm}^{-1}$ (b) $1.101 \times 10^{-2} \Omega^{-1} \text{ cm}^{-1}$ (c) $2.202 \times 10^{-3} \Omega^{-1} \text{ cm}^{-1}$ (d) $2.202 \times 10^3 \Omega^{-1} \text{ cm}^{-1}$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

84. [UNKNOWN] paper-ii

72. What will be the correct order of mobility of the following alkali metal ions in aqueous solution? (a) Li⁺ > Na⁺ > K⁺ > Rb⁺ (b) Na⁺ > K⁺ > Rb⁺ > Li⁺ (c) K⁺ > Rb⁺ > Na⁺ > Li⁺ (d) Rb⁺ > K⁺ > Na⁺ > Li⁺ ===== PAGE 15 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

85. [UNKNOWN] paper-ii

73. During electrolysis of silver nitrate solution, loss in the anode compartment was 0.589 g. In a Voltmeter connected in series, 1.259 g of silver was deposited. What would be the transport number of two ions? (a) t⁺ = 0.468 and t⁻ = 0.532 (b) t⁺ = 0.532 and t⁻ = 0.468 (c) t⁺ = 0.670 and t⁻ = 0.330 (d) t⁺ = 0.330 and t⁻ = 0.670

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

86. [UNKNOWN] paper-ii

74. The molar conductivities of KOAc and HCl at infinite dilution in water at 25°C are 90.0 and 426.2 S cm² mol⁻¹, respectively. To calculate the molar conductivity of HOAc at infinite dilution, the additional molar conductivity value of which one of the following electrolytes will be required? (a) NaCl (b) KCl (c) KNO₃ (d) NaOH

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

87. [UNKNOWN] paper-ii

75. What will be the effect of dilution on specific and equivalent conductance of an aqueous solution of a strong electrolyte? (a) The specific conductance decreases and equivalent conductance increases (b) The specific conductance increases and equivalent conductance decreases (c) Both will increase (d) Both will decrease

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

88. [UNKNOWN] paper-ii

76. The mean ionic activity co-efficient is denoted as $\gamma_{\pm}$, while that of cation and anion are denoted as γ_{+} and γ_{-} , respectively. The correct relation among them will be: (Given that v⁺ and v⁻ are the number of cations and the anions respectively in the molecular formula of the compound) (a) $\gamma_{\pm} = (\gamma_{+}^{v_{+}} \gamma_{-}^{v_{-}})^{1/(v_{+} + v_{-})}$ (b) $\gamma_{\pm} = (\gamma_{+}^{v_{+}} \gamma_{-}^{v_{-}})^{1/(v_{+} - v_{-})}$ (c) $\gamma_{\pm} = (\gamma_{+}^{v_{+}} \gamma_{-}^{v_{-}})^{1/(v_{+} \times v_{-})}$ (d) $\gamma_{\pm} = (\gamma_{+}^{v_{+}} \gamma_{-}^{v_{-}})^{1/(v_{+} + v_{-})}$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

89. [UNKNOWN] paper-ii

77. The specific conductance of 0.01 M solution of acetic acid was found to be 0.015 S m⁻¹ at 25°C. Molar conductance of acetic acid at infinite dilution is $400 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$. The degree of dissociation of acetic acid will be: (a) 0.375 (b) 0.0375 (c) 0.00375 (d) 0.357

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

90. [UNKNOWN] paper-ii

78. In a moving boundary experiment with 0.01 mole LiCl, the boundary in a tube with cross sectional area of 0.125 cm² moves through 8.1 cm in 1500 sec for a current of 1.8×10^{-3} ampere. What will be the value of transport number of Cl⁻? (Given that F = 96,500 Coulomb and temperature is 300K) (a) 0.639 (b) 0.361 (c) 0.510 (d) 0.369

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

91. [UNKNOWN] paper-ii

79. When HCl is titrated against NaOH, the conductance at the end point is: (a) Zero (b) Minimum (c) Maximum (d) Infinite

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

92. [UNKNOWN] paper-ii

80. A cell is bathed in a solution that has a greater osmolarity than that of cytosol. The solution can be described as: (a) Hypertonic (b) Hypotonic (c) Isotonic (d) Hydrotonic ===== PAGE 17 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

93. [UNKNOWN] paper-ii

81. Which cationic species is more stable in each of the following pairs? (a) 1 and 3 (b) 1 and 4 (c) 2 and 4 (d) 2 and 3

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

94. [UNKNOWN] paper-ii

82. Which one of the following structures is non-aromatic? (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

95. [UNKNOWN] paper-ii

83. The o/p-directing effect of methyl group in electrophilic substitution reaction of toluene is due to: (a) Inductive effect (b) Hyperconjugation effect (c) Resonance effect (d) Electromeric effect ===== PAGE 18 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

96. [UNKNOWN] paper-ii

84. The relative order of basicity of the following amines is: (a) $1 > 2 > 3 > 4$ (b) $4 > 1 > 2 > 3$ (c) $4 > 3 > 2 > 1$ (d) $2 > 3 > 1 > 4$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

97. [UNKNOWN] paper-ii

85. Which molecule in each of the following pairs has a higher dipole moment? (a) 1 and 3 (b) 1 and 4 (c) 2 and 3 (d) 2 and 4

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

98. [UNKNOWN] paper-ii

86. Which one of the following carbocations is NOT expected to rearrange? (a) (b) (c) (d) ===== PAGE 19 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

99. [UNKNOWN] paper-ii

87. Classify the following species as electrophiles (E) and nucleophiles (N) in organic reactions: SO_3 , Cl^+ , NO_2^+ , CH_3NH_2 , H_3O^+ , BH_3 , CN^- (a) $\text{E} = \text{SO}_3$, Cl^+ , NO_2^+ , H_3O^+ and $\text{N} = \text{BH}_3$, CH_3NH_2 , CN^- (b) $\text{E} = \text{Cl}^+$, NO_2^+ , H_3O^+ and $\text{N} = \text{SO}_3$, BH_3 , CH_3NH_2 , CN^- (c) $\text{E} = \text{SO}_3$, BH_3 , Cl^+ , NO_2^+ , H_3O^+ and $\text{N} = \text{CH}_3\text{NH}_2$, CN^- (d) $\text{E} = \text{CH}_3\text{NH}_2$, NO_2^+ , H_3O^+ , Cl^+ and $\text{N} = \text{SO}_3$, BH_3 , CN^-

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

100. [UNKNOWN] paper-ii

88. Which one of the following molecules does NOT exist as a meso isomer? (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

101. [UNKNOWN] paper-ii

89. Which one of the following molecules does NOT have a plane of symmetry? (a) (b) (c) (d) ===== PAGE 20 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

102. [UNKNOWN] paper-ii

90. Which one of the following amino acids is achiral? (a) Alanine (b) Glycine (c) Proline (d) Valine

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

103. [UNKNOWN] paper-ii

91. Consider the following compound [A]: Which one of the following represents an enantiomer of [A]? (a) (b) (c) (d)

===== PAGE 21 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

104. [UNKNOWN] paper-ii

92. The total number of optical isomers possible for cholesterol is: (a) 128 (b) 256 (c) 64 (d) 512

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

105. [UNKNOWN] paper-ii

93. Consider the following groups attached to a chiral carbon centre: What is the decreasing order of priority of above groups based on Cahn-Ingold-Prelog (CIP)- nomenclature? (a) $1 > 4 > 2 > 3$ (b) $1 > 2 > 3 > 4$ (c) $2 > 1 > 4 > 3$ (d) $4 > 1 > 3 > 2$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

106. [UNKNOWN] paper-ii

94. What is the correct configuration of C2 and C3 carbons in the following compound? (a) 2R, 3S (b) 2S, 3R (c) 2R, 3R (d) 2S, 3S

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

107. [UNKNOWN] paper-ii

95. The E/Z-nomenclature for the following molecules is: (a) 1: Z, 2: Z, 3: Z, 4: E (b) 1: Z, 2: E, 3: E, 4: E (c) 1: E, 2: Z, 3: Z, 4: E (d) 1: E, 2: Z, 3: E, 4: Z ===== PAGE 22 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

108. [UNKNOWN] paper-ii

96. The total number of stereoisomers possible for the given compound is: (a) 4 (b) 6 (c) 8 (d) 10

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

109. [UNKNOWN] paper-ii

97. Two compounds [X] and [Y] are given below: The relationship between [X] and [Y] is: (a) Identical (b) Diastereomers (c) Enantiomers (d) Constitutional isomers

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

110. [UNKNOWN] paper-ii

98. Which one of the following is the most stable conformation of ethylene glycol? (a) (b) (c) (d) ===== PAGE 23 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

111. [UNKNOWN] paper-ii

99. Consider the following compound: Which one of the following substrate sets is the best for synthesizing above compound? (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

112. [UNKNOWN] paper-ii

100. The major product(s) formed in the following reaction is/are: (a) (b) (c) (d) ===== PAGE 24 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

113. [UNKNOWN] paper-ii

101. The relative order of reactivity of the following chlorides in SN1 reaction is: (a) $2 > 1 > 3$ (b) $3 > 2 > 1$ (c) $1 > 2 > 3$ (d) $1 > 3 > 2$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

114. [UNKNOWN] paper-ii

102. The major product formed in the following reaction is: (a) (b) (c) (d) ===== PAGE 25 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

115. [UNKNOWN] paper-ii

103. The major product formed in the following reaction is: (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

116. [UNKNOWN] paper-ii

104. Which one of the following bromides CANNOT form an alkene by E2 process? (a) 1-Bromo-2,2-dimethylpropane (b) 2-Bromo-2-methylbutane (c) 2-Bromo-3-methylbutane (d) 1-Bromo-3-methylbutane

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

117. [UNKNOWN] paper-ii

105. The major product formed in the following reaction is: (a) (b) (c) (d) ===== PAGE 26 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

118. [UNKNOWN] paper-ii

106. The major product formed in the following reaction is: (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

119. [UNKNOWN] paper-ii

107. The major products [X] and [Y] formed in the following transformations are: (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

120. [UNKNOWN] paper-ii

108. The major product formed in the reaction of 3-phenylpropene with HBr in the presence of peroxide is: (a) 2-Bromo-1-phenylpropane (b) 1,2-Dibromo-3-phenylpropane (c) 1-Bromo-3-phenylpropane (d) 3-(p-Bromophenyl) propene ===== PAGE 27 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

121. [UNKNOWN] paper-ii

109. The major product formed in the following reaction is: (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

122. [UNKNOWN] paper-ii

110. The major product formed in the reaction of N,N-dimethylaniline with concentrated H₂SO₄/HNO₃ mixture is: (a) (b) (c) (d) ===== PAGE 28 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

123. [UNKNOWN] paper-ii

111. The major product formed in the following reaction is: (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

124. [UNKNOWN] paper-ii

112. The major product(s) formed in the following reaction is/are: (a) (b) (c) (d) ===== PAGE 29 =====

125. [UNKNOWN] paper-ii

113. Which one of the following reactions is NOT an example of ipso-substitution? (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

126. [UNKNOWN] paper-ii

114. The product formed by the reaction of toluene with chlorine in the presence of sunlight is: (a) o-chlorotoluene (b) 2,4-dichlorotoluene (c) p-chlorotoluene (d) benzylchloride ===== PAGE 30 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

127. [UNKNOWN] paper-ii

115. The major product formed in the following reaction is: (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

128. [UNKNOWN] paper-ii

116. The major product formed in the following reaction is: (a) (b) (c) (d) ===== PAGE 31 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

129. [UNKNOWN] paper-ii

117. The major product formed in the following reaction is: (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

130. [UNKNOWN] paper-ii

118. The major product formed in the following reaction is: (a) (b) (c) (d) ===== PAGE 32 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

131. [UNKNOWN] paper-ii

119. The major products [X] and [Y] formed in the following reactions are: (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf

132. [UNKNOWN] paper-ii

120. The relative order of the rate of acetolysis among the following norbornyl derivatives is: (a) $1 > 2 > 3$ (b) $1 > 3 > 2$ (c) $3 > 2 > 1$ (d) $3 > 1 > 2$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-CHEMISTRY.pdf